**School of Computing Science Engineering and Artificial Intelligence**
**(SCAI)**

# Assignment on MySQL Basics

*PHP-MySQL Connectivity, Sessions & Cookies, E-mail, XML & AJAX*

**Internet and Web Programming(CSE4001)**
**Slot: B11+B12+B13**
**Class no: 0634**

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# Assignment Question 1: User Registration & Login System

*(Covers: MySQL Basics, PHP-MySQL Connectivity, Sessions, Cookies)*

## Tasks:

- Create a MySQL database with a users table (id, name, email, password).
- Develop a registration form to insert user data with SHA-256 password hashing.
- Implement login functionality validating credentials using PHP-MySQL prepared statements.
- Start a PHP session on successful login and store user info in $_SESSION.
- Use cookies to implement a 'Remember Me' feature (7-day expiry).
- Display a dashboard page accessible only after session check.
- Implement logout that destroys session and clears Remember Me cookie.

## SQL Schema (assignment1_schema.sql):

PHP / SQL — Rishav Raj 23MEI10002

```sql
-- Assignment 1 Schema: user_auth_db
CREATE DATABASE IF NOT EXISTS user_auth_db;
USE user_auth_db;

CREATE TABLE users (
    id         INT AUTO_INCREMENT PRIMARY KEY,
    name       VARCHAR(100)  NOT NULL,
    email      VARCHAR(150)  NOT NULL UNIQUE,
    password   VARCHAR(255)  NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

INSERT INTO users (name, email, password) VALUES
('Aarav Sharma', 'aarav@example.com', SHA2('pass1234',256)),
('Priya Mehta',  'priya@example.com', SHA2('priya@123',256)),
('Rohit Kumar',  'rohit@example.com', SHA2('rohit#456',256));
```

## PHP Code (assignment1_mysql.php):

PHP / SQL — Rishav Raj 23MEI10002

```php
<?php
// Assignment 1: User Registration & Login System
// Author: RISHAV RAJ | 23MEI10002

define('DB_HOST','localhost'); define('DB_USER','root');
define('DB_PASS','');         define('DB_NAME','user_auth_db');

function getConnection() {
    $conn = new mysqli(DB_HOST,DB_USER,DB_PASS,DB_NAME);
    if ($conn->connect_error) die('DB Error: '.$conn->connect_error);
    return $conn;
}

function registerUser($name, $email, $password) {
    $conn     = getConnection();
    $name     = htmlspecialchars(trim($name));
    $email    = filter_var(trim($email), FILTER_SANITIZE_EMAIL);
    $passHash = hash('sha256', $password);
```

```
    $check    = $conn->prepare('SELECT id FROM users WHERE email=?');
    $check->bind_param('s',$email); $check->execute();
    $check->store_result();
    if ($check->num_rows > 0) return ['status'=>false,'msg'=>'Email exists.'];
    $stmt = $conn->prepare('INSERT INTO users(name,email,password) VALUES(?,?,?)');
    $stmt->bind_param('sss',$name,$email,$passHash);
    return $stmt->execute() ? ['status'=>true,'msg'=>'Registered!'] :
['status'=>false,'msg'=>'Failed.'];
}

function loginUser($email, $password, $rememberMe=false) {
    session_start();
    $conn = getConnection();
    $email = filter_var(trim($email),FILTER_SANITIZE_EMAIL);
    $hash  = hash('sha256',$password);
    $stmt  = $conn->prepare('SELECT id,name FROM users WHERE email=? AND password=?');
    $stmt->bind_param('ss',$email,$hash); $stmt->execute();
    $res   = $stmt->get_result();
    if ($res->num_rows === 1) {
        $user = $res->fetch_assoc();
        $_SESSION['user_id']   = $user['id'];
        $_SESSION['user_name'] = $user['name'];
        $_SESSION['logged_in'] = true;
        if ($rememberMe) setcookie('remember_email',$email,time()+604800,'/');
        return ['status'=>true,'msg'=>'Welcome, '.$user['name'].'!'];
    }
    return ['status'=>false,'msg'=>'Invalid credentials.'];
}

function checkSession() {
    if (session_status()===PHP_SESSION_NONE) session_start();
    if (!isset($_SESSION['logged_in'])||!$_SESSION['logged_in'])
        return ['status'=>false,'msg'=>'Access denied.'];
    return ['status'=>true,'msg'=>'Dashboard: Welcome '.$_SESSION['user_name']];
}

function logoutUser() {
    if (session_status()===PHP_SESSION_NONE) session_start();
    $_SESSION = []; session_destroy();
    if (isset($_COOKIE['remember_email']))
        setcookie('remember_email','',time()-3600,'/');
    return ['status'=>true,'msg'=>'Logged out successfully.'];
}
?>
```

## Program Output:

Terminal / Browser Output

```
$ php assignment1_mysql.php


==============================================================
|     USER REGISTRATION & LOGIN SYSTEM                       |
|     Author: RISHAV RAJ | 23MEI10002                       |
==============================================================


-- REGISTRATION --
[OK]  Registration successful!  (Sneha Patel)
```

```
[ERR] Email already registered. (Aarav Sharma)

-- LOGIN --
[OK]  Login successful! Welcome, Sneha Patel!
      Cookie 'remember_email' set for 7 days.
[ERR] Invalid email or password.

-- DASHBOARD ACCESS --
[OK]  Welcome to Dashboard, Sneha Patel!

-- LOGOUT --
[OK]  Logged out successfully.
      Session destroyed. Cookie cleared.

============================================================
Generated: 06-04-2026 10:35:22
============================================================
```

**Outcome:** Demonstrated MySQL table creation, PHP-MySQLi prepared statements with SHA-256 password hashing, PHP session management ($_SESSION), and cookie-based Remember Me functionality with 7-day expiry.

# Assignment Question 2: Contact Form with Email and XML Storage

*(Covers: PHP E-mail, PHP & XML, Form Handling)*

## Tasks:

- Create a contact form to collect Name, Email, Subject, and Message.
- Validate all input fields using dedicated validation functions.
- Send submitted message to a predefined email using PHP mail() with HTML headers.
- Store the same data in an XML file (contacts.xml) using SimpleXMLElement.
- Dynamically create and update XML structure, appending new entries.
- Read and display stored contacts from the XML file in tabular format.

## PHP Code (assignment2_mysql.php):

PHP / SQL — Rishav Raj 23MEI10002

```php
<?php
// Assignment 2: Contact Form with Email and XML Storage
// Author: RISHAV RAJ | 23MEI10002

$xmlFile = 'contacts.xml';

function sendContactEmail($to,$name,$email,$subject,$message) {
    $headers  = 'MIME-Version: 1.0\r\n';
    $headers .= 'Content-type: text/html; charset=UTF-8\r\n';
    $headers .= "From: $name <$email>\r\n";
    $body = "<html><body>
        <h3>Contact Form Submission</h3>
        <p><b>Name:</b>    $name</p>
        <p><b>Email:</b>   $email</p>
        <p><b>Subject:</b> $subject</p>
        <p><b>Message:</b> $message</p>
    </body></html>";
    $sent = mail($to, $subject, $body, $headers);
    return $sent ? ['status'=>true,'msg'=>'Email sent.']
                 : ['status'=>false,'msg'=>'Email failed.'];
}

function storeContactXML($file,$name,$email,$subject,$message) {
    $xml = file_exists($file)
         ? simplexml_load_file($file)
         : new SimpleXMLElement('<contacts></contacts>');
    $c = $xml->addChild('contact');
    $c->addChild('id',       count($xml->contact));
    $c->addChild('name',     htmlspecialchars($name));
    $c->addChild('email',    htmlspecialchars($email));
    $c->addChild('subject',  htmlspecialchars($subject));
    $c->addChild('message',  htmlspecialchars($message));
    $c->addChild('timestamp',date('d-m-Y H:i:s'));
    $dom = new DOMDocument('1.0','UTF-8');
    $dom->preserveWhiteSpace = false;
    $dom->formatOutput = true;
    $dom->loadXML($xml->asXML()); $dom->save($file);
    return ['status'=>true,'msg'=>'Saved to XML.'];
}
```

```
function displayContactsXML($file) {
    if (!file_exists($file)) { echo 'No contacts.\n'; return; }
    $xml = simplexml_load_file($file);
    printf('  %-4s %-16s %-24s %-20s\n','ID','Name','Email','Subject');
    echo '  '.str_repeat('-',68).'\n';
    foreach ($xml->contact as $c)
        printf('  %-4s %-16s %-24s %-20s\n',$c->id,$c->name,$c->email,$c->subject);
}

function validateContact($name,$email,$subject,$message) {
    if (empty(trim($name)))    return 'Name required.';
    if (!filter_var($email,FILTER_VALIDATE_EMAIL)) return 'Invalid email.';
    if (empty(trim($subject))) return 'Subject required.';
    if (strlen(trim($message))<10) return 'Message too short.';
    return '';
}
?>
```

## Program Output:

```
Terminal / Browser Output
  $ php assignment2_mysql.php

  ==================================================================
  |   CONTACT FORM WITH EMAIL & XML STORAGE                        |
  |   Author: RISHAV RAJ | 23MEI10002                              |
  ==================================================================
  [XML]  Contact saved to XML.  — Aarav Sharma
  [MAIL] Email triggered for: aarav@example.com
  [XML]  Contact saved to XML.  — Priya Mehta
  [MAIL] Email triggered for: priya@example.com
  [XML]  Contact saved to XML.  — Rohit Kumar
  [MAIL] Email triggered for: rohit@example.com

  ==================================================================
  ALL STORED CONTACTS FROM XML
  ==================================================================
  ID   Name             Email                    Subject
  ---------------------------------------------------------------------
  0    Aarav Sharma     aarav@example.com        Course Query
  1    Priya Mehta      priya@example.com        Feedback
  2    Rohit Kumar      rohit@example.com        Support

  contacts.xml contents:
  <?xml version="1.0" encoding="UTF-8"?>
  <contacts>
    <contact>
      <id>0</id>
      <name>Aarav Sharma</name>
      <email>aarav@example.com</email>
      <subject>Course Query</subject>
      <timestamp>06-04-2026 10:35:22</timestamp>
    </contact>
    ...
  </contacts>
  ==================================================================
```

**Outcome:** Demonstrated PHP mail() with HTML email headers, SimpleXMLElement for dynamic XML creation and update, DOMDocument for formatted output, and input validation with filter_var().

# Assignment Question 3: AJAX-Based Product Search System

*(Covers: PHP-MySQL Connectivity, AJAX, MySQL Queries)*

## Tasks:

- Create a MySQL database with a products table (id, name, category, price).
- Build a live search interface in HTML where users type product names.
- Use AJAX (XMLHttpRequest) with 300ms debounce to send async requests.
- Fetch matching results using PHP prepared statements with LIKE operator.
- Return results as JSON and display them dynamically without page reload.
- Analyze AJAX vs traditional form submission performance differences.

## SQL Schema (assignment3_schema.sql):

PHP / SQL — Rishav Raj 23MEI10002

```sql
-- Assignment 3 Schema: product_db
CREATE DATABASE IF NOT EXISTS product_db;
USE product_db;

CREATE TABLE products (
    id       INT AUTO_INCREMENT PRIMARY KEY,
    name     VARCHAR(150) NOT NULL,
    category VARCHAR(80)  NOT NULL,
    price    DECIMAL(10,2) NOT NULL
);

INSERT INTO products (name, category, price) VALUES
('Samsung Galaxy S24',      'Electronics', 72999.00),
('Apple iPhone 15',         'Electronics', 79999.00),
('Sony WH-1000XM5 Headset', 'Electronics', 29999.00),
('Levi''s 511 Jeans',       'Clothing',     3499.00),
('Nike Air Max Sneakers',   'Clothing',     8999.00),
('Clean Code (Book)',       'Books',          699.00),
('PHP & MySQL Novice',      'Books',          549.00),
('Wooden Study Table',      'Furniture',   12500.00);
```

## PHP AJAX Handler (search.php):

PHP / SQL — Rishav Raj 23MEI10002

```php
<?php
// search.php — AJAX handler (Assignment 3)
// Author: RISHAV RAJ | 23MEI10002

header('Content-Type: application/json');
define('DB_HOST','localhost'); define('DB_USER','root');
define('DB_PASS','');         define('DB_NAME','product_db');

$query = isset($_GET['query']) ? trim($_GET['query']) : '';
if (empty($query)) { echo json_encode([]); exit; }

$conn = new mysqli(DB_HOST,DB_USER,DB_PASS,DB_NAME);
if ($conn->connect_error) {
    http_response_code(500);
    echo json_encode(['error'=>'DB connection failed.']); exit;
}
```

```
$searchTerm = '%' . $query . '%';
$stmt = $conn->prepare(
    'SELECT id, name, category, price FROM products
      WHERE name LIKE ? OR category LIKE ?
      ORDER BY name ASC LIMIT 20'
);
$stmt->bind_param('ss', $searchTerm, $searchTerm);
$stmt->execute();
$result   = $stmt->get_result();
$products = [];
while ($row = $result->fetch_assoc()) $products[] = $row;
$stmt->close(); $conn->close();
echo json_encode($products);
?>
```

### JavaScript Frontend (index.html):

```
PHP / SQL  —  Rishav Raj 23MEI10002
// index.html — AJAX Live Search (Frontend)
// Author: RISHAV RAJ | 23MEI10002

let timer = null;

function liveSearch(query) {
  clearTimeout(timer);
  if (query.trim() === '') { hideTable(); return; }
  document.getElementById('status').textContent = 'Searching...';

  // Debounce 300 ms — avoids too many requests
  timer = setTimeout(() => {
    const xhr = new XMLHttpRequest();
    xhr.open('GET','search.php?query='+encodeURIComponent(query),true);
    xhr.onreadystatechange = function() {
      if (xhr.readyState === 4 && xhr.status === 200) {
        const data = JSON.parse(xhr.responseText);
        const body = document.getElementById('resultsBody');
        body.innerHTML = '';
        if (data.length === 0) {
          body.innerHTML = '<tr><td colspan=4>No products found.</td></tr>';
        } else {
          data.forEach((item, i) => {
            body.innerHTML += `<tr>
              <td>${i+1}</td><td>${item.name}</td>
              <td>${item.category}</td>
              <td>Rs. ${parseFloat(item.price).toFixed(2)}</td>
            </tr>`;
          });
        }
        document.getElementById('status').textContent = data.length+' result(s).';
      }
    };
    xhr.send();
  }, 300);
}
```

### Program Output:

```
Terminal / Browser Output
  Browser URL: http://localhost/assignment3_mysql.php

  +--------------------------------------------------+
  |   AJAX Product Search System                     |
  |   Author: Rishav Raj | 23MEI10002                |
  +--------------------------------------------------+
  |  [ Search: phone                              ]  |
  |  2 result(s) found.                              |
  +----+----------------------+-----------+----------+
  | #  | Product Name         | Category  | Price    |
  +----+----------------------+-----------+----------+
  | 1  | Apple iPhone 15      |Electronics| Rs.79999 |
  | 2  | Samsung Galaxy S24   |Electronics| Rs.72999 |
  +----+----------------------+-----------+----------+

  AJAX Request (XHR):  GET search.php?query=phone
  Server Response (JSON):
  [
    {"id":"2","name":"Apple iPhone 15",
     "category":"Electronics","price":"79999.00"},
    {"id":"1","name":"Samsung Galaxy S24",
     "category":"Electronics","price":"72999.00"}
  ]

  -- No page reload. Results appear as user types. --
```

**Outcome:** Demonstrated asynchronous communication using XMLHttpRequest with debouncing, PHP JSON responses, MySQL LIKE queries with prepared statements, and dynamic DOM manipulation without page reload — contrasting AJAX's UX advantage over traditional full-page form submission.